

AASHISH SINHA

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EDUCATION

Indian Institute of Technology (ISM) Dhanbad

Bachelor of Technology in Mechanical Engineering

Dhanbad, India

Nov. 2022 – May 2026

The Senior Study School

High School, Science

Amritsar, India

March 2022

EXPERIENCE

RRC: IIIT Hyderabad

Research Intern: Under Prof. Spandan Roy

Feb 2026 – Present

Hyderabad, India

- **Project:** Graph-Informed MPC for Drone Swarms
 - Designed a MPC framework for collaborative multi-drone payload transport.
 - V1: Trained a GNN to predict nonlinear MPC residuals.
 - V2: Predicted tension forces acting on each drone in the swarm.
 - Generalized the model for variable swarm size/payload weights.
- **Project:** ZeroTrack: Multi-agent RL policy for Cooperative Aerial Transport (Ongoing)
 - Set-up parallelized quadrotor simulation in MuJoCo/MJX.
 - Built RL policy to output direct motor thrusts, bypassing standard PID controls.
 - V1: Trained baseline IPPO policy for payload tracking for $N = 2$ quadrotor swarm size.
 - V2: Trained MAPPO policy for $N \geq 2$, variable quadrotor/payload masses and cable length.
- **Project:** Vision-Guided Aerial Manipulator - Grabbing and Placing
 - Implemented minimum-snap for smooth quadrotor trajectory.
 - Integrated motion controllers with VLM outputs to execute flight path.
 - Worked on grabbing and placing task with the manipulator on the quadrotor.

Robotics Lab: IIT Dhanbad

Guided by Prof. Arun Dayal Udai

Nov 2025

Dhanbad, India

- **Project:** Humanoid - Grasping with RGB-D Camera
 - Processed Intel-RealSense (RGB-D) data for 3D pose estimation of an object.
 - Solved robot IK and executed bottle-grasping task.
- **Project:** Cart-Pole Stabilization System
 - V1: Implemented an energy-based control algorithm & LQR balancing controller.
 - V2: Trained PPO algorithm.

Automation Lab: IIT Dhanbad

Research Assistant: Under Prof. Vivek Bajpai

Aug 2025 – Apr 2026

Dhanbad, India

- **Project:** Vision-Guided Mechatronic Gantry System
 - Designed and manufactured a 3-DOF mechatronic gantry crane & parallelogram gripper(3D printed).
 - Mapped 2D camera feeds into 3D world frame using perspective transforms.
 - Deployed a fine-tuned, YOLO model on RPi for steel coil detection.

DG Takano

Mechanical Research Intern

May 2025 – Jul 2025

Osaka, Japan

- **Project: Precision Mechanical Actuation & Structural Design**
 - Engineered a 3-position mechanical switch system for variable fluid control(sustainability).
 - Increased the magnetic coupling force by 300%.
 - Designed for manufacturing & reduced production costs by 12,000 JPY/unit.

NVCTI (Institute Innovation Hub)

Mentored by Dr. Ashish Siddharth

Jan 2025 – Apr 2025

Dhanbad, India

- **Project:** LLM-Integrated Autonomous Mobile Robot (AMR)
 - Designed a differential-drive AMR in Fusion.
 - Deployed SLAM/Nav2 on RPi4 using ROS2.
 - Components: 2D LiDAR, IMU, magnetometer, EKF for fusion, DC motor with encoders.
 - Pipeline is Voice > Gemini API > Navigation.

HACKATHON/PROJECTS

Multi-Task Diffusion Policy | *PyTorch, Diffusion Models, Multi-Task Learning*

- Built a unified multi-task PyTorch Diffusion Policy to solve dual robotic manipulation tasks.
- Custom projection heads to align disparate observation space.

World Model for Autonomous Racing | *PyTorch, VAE, CMA-ES*

- Built a World Model with a VAE/LSTM-MDN memory to predict visual states in CarRacing-v3
- Optimized controller weights using CMA-ES. Controller maps latent states to driving actions.

GAN | *PyTorch, Generator/Discriminator*

- Trained a GAN on 60K anime faces.
- Fine-tuned the model for realism in facial features.

TECHNICAL SKILLS

Programming: Python/C++, Linux, Git, Docker, REST APIs (FastAPI/Flask)

Robotics & Simulation: ROS2, MuJoCo, PX4, Gazebo

AI & Machine Learning: PyTorch, RL/PPO, GNN, CNN, GAN, World Model

Control & Perception: PID/MPC, YOLO, SLAM/Nav2, LiDAR

Hardware & Prototyping: RPi/ESP32, IMU, 3D Printing, CAD, DFM

Robots: Svaya (Humanoid), Yaskawa 6-DOF serial

ACHIEVEMENTS

- **Regional Finalist: NXP 2025** Autonomous ROS2/Gazebo warehouse navigation and CV.
- **Stage 2 Qualifier: e-Yantra 2025 (IIT Bombay)** Robotic arm manipulation for agriculture robot.

RELEVANT COURSEWORK

Robotics & Control: Kinematics & Dynamics of Machinery, PID/MPC, Mechatronics

AI & Computing: Computer Vision, Reinforcement Learning, Data Structures & Algorithms, Embedded Systems

Mathematics: Linear Algebra, Optimization Techniques, Probability & Statistics